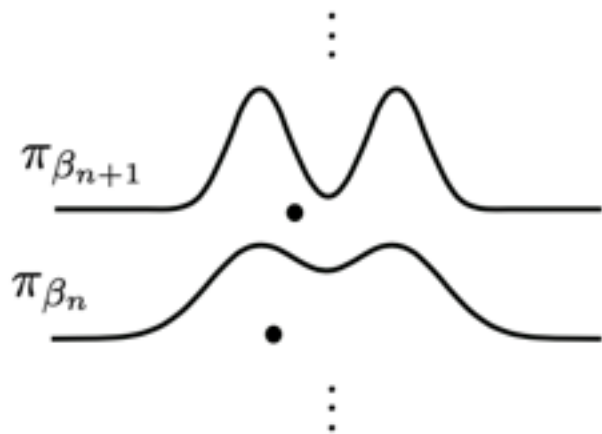
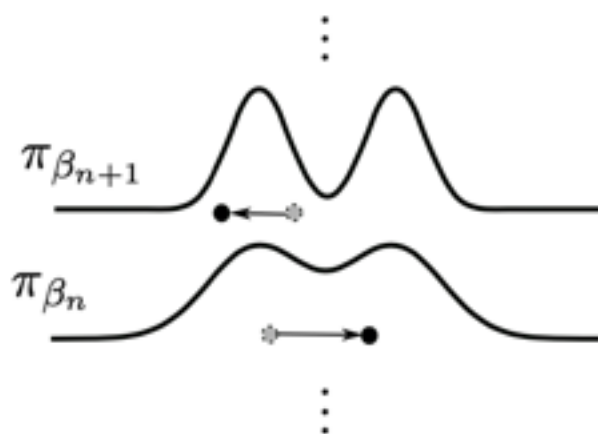


Initialize



Local Exploration



1

2

► **Exploration:** Update components of $\mathbf{x} = (x^0, \dots, x^N)$ in parallel using MCMC

Assume that an independent sample from reference

► Update the x^n using the local move K_{β_n} targeting π_{β_n}

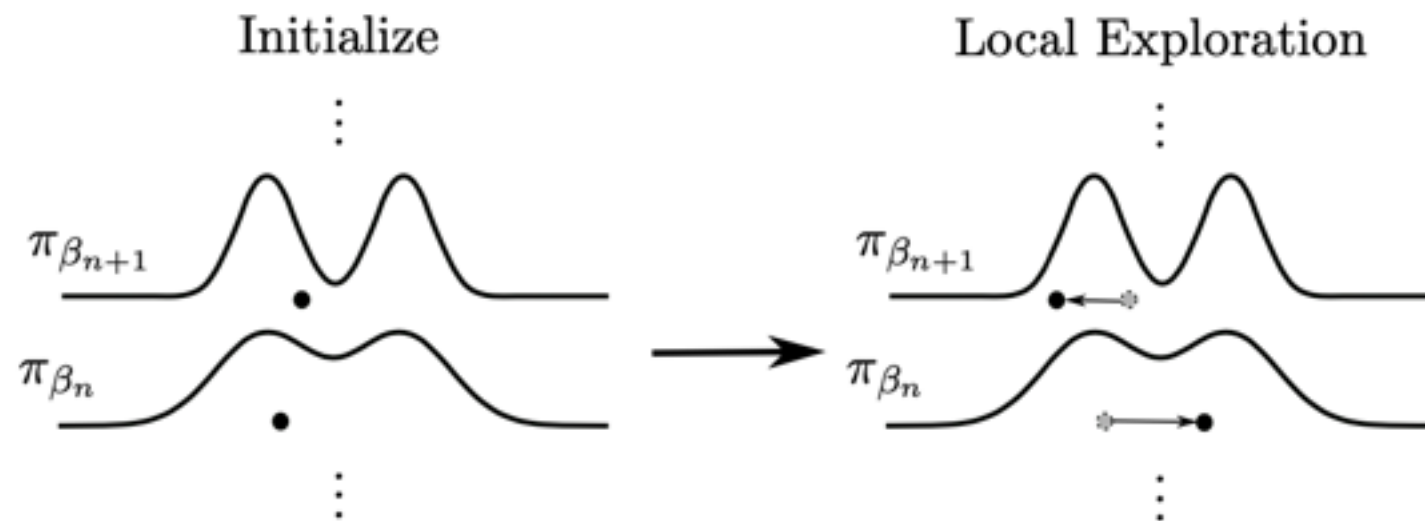
For example, Metropolis-Hastings, Gibbs sampling, RMM, MALA, HMC, etc.



$$K_{\text{expl}}(\mathbf{x}, d\mathbf{x}) = K_{\beta_0} \otimes \cdots \otimes K_{\beta_N}(\mathbf{x}, d\mathbf{x})$$

$$= \eta(\mathrm{d}x^{'0}) \prod_{n=1}^N K_{\beta_n}(x^n, \mathrm{d}x^{'n})$$

LOCAL EXPLORATION



- ▶ **Exploration:** Update components of $\mathbf{x} = (x^0, \dots, x^N)$ in parallel using MCMC
- ▶ Update the x^n using the local move K_{β_n} targeting π_{β_n}
 - ▶ For example, Metropolis-Hastings, Gibbs sampling, RWM, MALA, HMC, etc.
 - ▶ Assume that an independent sample from reference

$$\begin{aligned}
 K_{\text{expl}}(\mathbf{x}, d\mathbf{x}') &= K_{\beta_0} \otimes \dots \otimes K_{\beta_N}(\mathbf{x}, d\mathbf{x}') \\
 &= \eta(dx'^0) \prod_{n=1}^N K_{\beta_n}(x^n, dx'^n)
 \end{aligned}$$

SWAP MOVE

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